

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – PSEUDOCODE WRITING TASK

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO5 – Naturalization (BL4, BL6)	Assessment:	Assessment Tool 1 – Pseudocode Writing Task
Weightage:	35% of CIA	Total Marks:	100
CO5 Statement:	Construct MST using shortest path algorithms and traverse the graph to model and analyse real-world problem.		
Student Name:	SUMETHA SRI	Reg. No.:	192521097
Section / Batch:		Date:	31/08/2026

Code of Conduct

I _____ Sumetha Sri.S/192521097 _____ (Name / Reg No)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- This test contains 5 Pseudocode Writing Task questions. Total: 100 Marks.
- When a student answer using pseudocode writing task should evaluate based on the ability to design clear, logical, and efficient pseudocode solutions for data structure problems.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Minimum Spanning Tree	Kruskal's Algorithm	1	20
Graph Sorting	Topological Sort	2	20
Shortest Path Algorithm	MST & Dijkstra's Algorithm	3	20
Shortest Path Algorithm	Dijkstra's Algorithm	4	20
Minimum Spanning Tree	Prim's or Kruskal's Algorithm	5	20
GRAND TOTAL:			100 Marks

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20	Demonstrates complete and clear understanding; handles edge cases effectively	Good understanding; minor gaps in edge case handling	Partial understanding; misses some requirements	Misinterprets problem; major gaps
Code Correctness	30	Fully correct; passes all test cases	Mostly correct; minor errors	Partially correct; several test cases fail	Incorrect or non-functional code
Code Quality & Readability	20	Clean, modular, well-commented, follows standards	Readable with some structure	Limited readability; poor structure	Unorganized and hard to read
Complexity Analysis	20	Accurate time & space complexity with justification	Correct but limited explanation	Incomplete or partially correct	Missing or incorrect
Testing & Validation	10	Thorough testing including edge cases	Adequate testing with few edge cases	Minimal testing	No testing evidence

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

CO5_AT1

PSEUDOCODE WRITING

1.

The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS Engineering logo, the lab name, and the user's name (SUMETHA SRI S) with a dropdown arrow. The main title is "PSEUDOCODE WRITING TASK". On the left, a "QUESTIONS" sidebar lists five questions, each with a green checkmark and "20 marks". Question 1 is selected. The main area displays the question details: "Q1 Minimum Spanning Tree - Kruskal's Algorithm" (20 marks). The question text asks for pseudocode for constructing an MST using Kruskal's Algorithm, with constraints: a. All zones must be connected, b. Total installation cost must be minimized, c. No redundant connections (no cycles) allowed, d. The network must remain fully connected. Below the question is a "Your Answer" text area containing the following pseudocode:

```
ELSE IF rank[u] > rank[v] THEN
    parent[v] ← u
ELSE
    parent[v] ← u
    rank[u] ← rank[u] + 1
```

At the bottom of the answer area are "Save Answer" and "Next →" buttons. A "5/5 answered" status is shown at the bottom left.

2.

The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS Engineering logo, the lab name, and the user's name (SUMETHA SRI S) with a dropdown arrow. The main title is "PSEUDOCODE WRITING TASK". On the left, a "QUESTIONS" sidebar lists five questions, each with a green checkmark and "20 marks". Question 2 is selected. The main area displays the question details: "Q2 Graph - Topological Sort" (20 marks). The question text asks for pseudocode for Topological Sort using DFS with cycle detection, with constraints: a. Detect cycle before sorting, b. Use DFS approach. Below the question is a "Your Answer" text area containing the following pseudocode:

```
TOPOLOGICAL_SORT(G = (V, E))

Input:
V = Set of vertices (tasks)
E = Set of directed edges (dependencies)
```

At the bottom of the answer area are "Save Answer" and "Next →" buttons. A "5/5 answered" status is shown at the bottom left.

3.



SIMATS Online Programming Lab
DS PROGRAMMING

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Q1
Minimum Spanning Tree - Kruskal's Algorithm
20 marks

Q2
Graph - Topological Sort
20 marks

Q3
Shortest Path Algorithm - Minimum Spanning Tree & Dijkstra's
20 marks

Q4
Shortest Path Algorithm - Dijkstra's Algorithm
20 marks

Q5
Minimum Spanning Tree - Prim's or Kruskal's Algorithm
20 marks

5/5 answered

Q3 Shortest Path Algorithm - Minimum Spanning Tree & Dijkstra's
20 marks

Write pseudocode integrating MST + Dijkstra for Smart transportation system.
Constraints:
a. Use MST for infrastructure
b. Use shortest path for routing

Your Answer

SMART_TRANSPORTATION_SYSTEM($G = (V, E)$)

Input:
V = Set of locations
E = Set of roads/connections
Each edge (u, v, w) has a cost/distance w

Save Answer
Next →

4.



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PSEUDOCODE WRITING TASK
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Q1
Minimum Spanning Tree - Kruskal's Algorithm
20 marks

Q2
Graph - Topological Sort
20 marks

Q3
Shortest Path Algorithm - Minimum Spanning Tree & Dijkstra's
20 marks

Q4
Shortest Path Algorithm - Dijkstra's Algorithm
20 marks

Q5
Minimum Spanning Tree - Prim's or Kruskal's Algorithm
20 marks

5/5 answered

Q4 Shortest Path Algorithm - Dijkstra's Algorithm
20 marks

Shortest Path with "Toll-Free" Voucher: You have a voucher that allows you to traverse one edge for free (weight = 0). Write pseudocode to find the shortest path from S to D utilizing this voucher optimally.

Your Answer

SHORTEST_PATH_WITH_VOUCHER(G, S, D)

Input:
G = Graph with vertices V and edges E
S = Source vertex
D = Destination vertex

Save Answer
Next →

5.

PSEUDOCODE WRITING TASK

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QUESTIONS

Q1 ✓
Minimum Spanning Tree -
Kruskal's Algorithm
20 marks

Q2 ✓
Graph - Topological Sort
20 marks

Q3 ✓
Shortest Path Algorithm -
Minimum Spanning Tree &
Dijkstra's
20 marks

Q4 ✓
Shortest Path Algorithm -
Dijkstra's Algorithm
20 marks

Q5 ✓
Minimum Spanning Tree -
Prim's or Kruskal's Algorithm
20 marks

5/5 answered

Q5 Minimum Spanning Tree - Prim's or Kruskal's Algorithm

20 marks

Second Best Spanning Tree: Design pseudocode to find the "Second Best" MST.
Hint: Find the MST first, then for every edge in the MST, try removing it and finding the next best edge to replace it.

Your Answer

SECOND_BEST_MST($G = (V, E)$)

Input:

V = Set of vertices

E = Set of weighted edges

Save Answer